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Superseding
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MILITARY SPECIFICATION

HOSE, RUBBER, AIRCRAFT PAINT FINISH REMOVER

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one type of hose for handling aircraft paint finish remover.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Federal

ZZ-R-601	Rubber-Goods; General Specifications (Methods of Physical Tests and Chemical Analyses)

Military

MIL-A-6091	Alcohol; Ethyl, Specially Denatured, Aircraft
JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-139	Packaging and Packing for Overseas Shipment - Plywood, Container Grade

U. S. Air Force

14119	Remover; Paint, Aircraft, Finish

2.1.2 Other Publications.-

Military Standards (Book)

MIL-STD-129	Marking of Shipments

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained by the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials.- All materials which are not specifically described herein shall be of the best quality and entirely suitable for the purpose intended.

3.2 Construction.- The hose shall consist of a seamless compounded inner tube, reinforcement, and compounded cover. The bore shall be clean and free from pitting as is consistent with good manufacturing practice.

3.3 Diameter.- The hose shall be furnished in the following sizes as specified by the procuring agency:

Inside Diameter Inches ±1/32	Outside Diameter Inches ±1/32
3/8	3/4
1/2	7/8
1	1-3/8

3.4 Length.- The maximum acceptable length of the hose shall be 50 feet; the minimum length shall be that specified.

3.5 Physical Properties.-

3.5.1 Adhesion Between Tube and Adjacent Parts.- The average tension required to cause separation between tube and adjacent parts at the specified rate shall be at least 10 pounds per inch for unaged hose; after aging for 72 hours at 70° ±5°F in aircraft paint finish remover, the average tension required to cause separation between tube and adjacent parts shall be at least 5 pounds per inch.

3.5.2 Hydrostatic Burst Pressure.- The hose shall not leak or show any imperfections when subjected to the hydrostatic pressure of 500 psi when applied for not less than 30 seconds.

3.5.3 Low Temperature Flexibility.- The hose shall withstand bending at a temperature of -55° ±2°F without failure.

3.5.4 Water and Alcohol Resistance.- The tensile change of the tube and cover materials when tested as specified in paragraph 4.4.4 shall not exceed 35 percent.

3.5.5 Tensile Strength.- The tensile strength of the tube and cover shall be not less than 800 psi.

3.5.6 Ultimate Elongation.- The ultimate elongation of the tube and cover shall be not less than 250 percent.

3.5.7 Volume Change.- The volume change of the tube and cover specimens after immersion for 72 hours at 75° ±5°F in aircraft paint finish remover shall not exceed 45 percent.

3.6 Identification of Product.- Each piece or length of hose shall have two red stripes along its length or spirally around the cover. The stripes may be colored gum inlaid on the cover, or gasoline and oil-resistant lacquer of the proper color. In addition, each length of hose greater than 10 feet shall have a label inlaid in the cover and vulcanized hereto. The label shall furnish the following information:

Specification MIL-H-6439A
Date Manufactured (Month and Year)
Manufacturer's Name or Trade-Mark

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3.7 Workmanship.- The hose shall be uniform in quality and free from defects; all details of workmanship shall be in accordance with high-grade commercial practice covering this type of work.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- All the tests required for the testing of hose are classified as Inspection tests, for which necessary sampling techniques and methods of testing are specified in this section. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.2 Test Conditions.- Unless otherwise specified for the particular test, all tests shall be accomplished in accordance with Specification ZZ-R-601.

4.3 Examination of Product.- All hose shall be carefully examined in order to insure conformance to the requirements of this specification with respect to materials, workmanship, size, construction, and marking.

4.4 Sampling Tests.- A 4-foot length of hose shall be selected at random for test purposes from each 1,000 feet or less. When short lengths are specified, the manufacturer shall furnish a 4-foot length of hose from the same lot for test. Specimens required for testing shall be in addition to the quantity specified in the order and shall be furnished free of charge to the procuring agency. Samples of hose selected shall be subjected to the tests which follow.

4.4.1 Adhesion.- Adhesion specimens shall be prepared as described in Specification ZZ-R-601. Where immersion is specified, a section of the hose approximately 4 inches long shall be immersed and the test specimens prepared from the center portion of this section after immersion. When tested, hose shall meet the requirements specified in paragraph 3.5.1.

4.4.2 Burst Test.- For the Burst test, pressures shall be applied by any convenient means at the rate of approximately 2,000 ±1,000 psi per minute.

4.4.3 Low Temperature Flexibility.- A straight piece of hose at least 12 inches long shall be conditioned at -55° ±2°F for 5 hours, and within 2 seconds the hose shall be flexed around a cooled mandrel or cylinder 10 times the outside diameter of the hose.

4.4.4 Water and Alcohol Resistance.- The tensile loss of the tube in 50-percent alcohol conforming to Specification MIL-A-6091, and 50-percent water at a temperature of 158°F for 24 hours under reflux shall not exceed 35 percent. The samples shall be tested as outlined in Specification ZZ-R-601 except the tensile strength shall be taken within 15 minutes after removal from the solution. The tensile strength calculations shall be based upon the original cross sectional area of the test specimens.

4.4.5 Tensile and Elongation Tests.- The test specimens used for obtaining tensile strength and elongation shall be 1/4 by 2 inches, type B as covered by Specification ZZ-R-601 and shall be cut from hose samples. The specimens shall be buffed to a uniform thickness. Hose, when tested, shall meet the requirements specified in paragraphs 3.5.5 and 3.5.6.

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4.4.6 Volume Change Test.- Volume change specimens, 1 by 2 inches or by the circumference of the tube (whichever is the smaller) shall be cut from the tube. The weight of water displaced by the tube shall be determined before and after immersion in aircraft paint finish remover conforming to Specification 14119. A jolly or other suitable balance shall be used. The displacement after immersion in test fluids minus the initial displacement multiplied by 100 and divided by the initial displacement shall be considered the percentage change in volume.

4.5 Rejection and Retest.- When samples of hose selected as representative of a day's run fail to meet the requirements of this specification, the day's run shall be rejected. Hose which has been rejected may be replaced or reworked to correct the defects, retested, and resubmitted for acceptance; it shall not be resubmitted, however, without furnishing full particulars concerning previous rejection and corrective measures taken.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging.- Unless otherwise specified, all hose shall be packaged and shipped in a straight 10-foot length. Longer lengths, if specified, may be coiled provided the inside diameter of the coil is not less than 2 feet. Hose packaged in a given carton shall be of the same sizes and lengths whenever practicable.

5.3 Packing.- Unless otherwise specified, all items shall be packed for domestic shipment. Each shipping container, insofar as possible, shall contain an identical number of articles, shall be of a uniform size, and shall be designed to inclose the contents in a snug, tight-fitting manner. The gross weight of the shipping container when packed for shipment shall not exceed approximately 200 pounds.

5.3.1 Domestic Packing.- Unless otherwise specified, the hose shall be packed in substantial commercial cleated plywood boxes, nailed wood boxes, fiberboard boxes, or in suitable kraft paper and burlap, wrapped bundles. The container shall be so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Except as specified herein, the shipping containers shall conform to the requirements of Consolidated Freight Classification Rules in effect at the time of shipment, except that the fiberboard container shall have a minimum Mullen test of 275 pounds, and shall have a taped inner liner of like material. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.3.2 Overseas Packing.- Each reel of hose shall be packed in a shipping container in accordance with Specification JAN-P-105, except that the plywood shall conform to Specification JAN-P-139, type B, condition I, or Specification JAN-P-106. The reel shall be fastened in the center of the bottom of the shipping container by means of four carriage bolts, lock washers, and nuts. The heads of the bolts shall be on the outside of the container. The straight ends of the hose shall be packed lengthwise in a shipping container in accordance with Specification JAN-P-105 or JAN-P-106.

5.4 Marking and Labeling.-

5.4.1 Packages.- Each package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the packages are opened:

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HOSE, RUBBER, AIRCRAFT PAINT FINISH REMOVER
Specification MIL-H-6439A
Inside Diameter
Quantity Contained
Stock No.
Contract or Order No.
Name of Manufacturer
Name of Contractor (if different from manufacturer)
Date of Manufacture (month and year)

5.5.2 Shipping Containers.- Each shipping container shall be marked in accordance with the requirements applicable to the individual Services, as specified in MIL-STD-129.

6. NOTES

6.1 Intended Use.- The hose covered by this specification is intended for use in aircraft paint finish remover lines for spray equipment.

6.2 Ordering Data.- Requisitions, contracts, and orders should state size and length to be furnished, and whether overseas packing is desired.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Air Force

Other Interest:
Army - Q

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